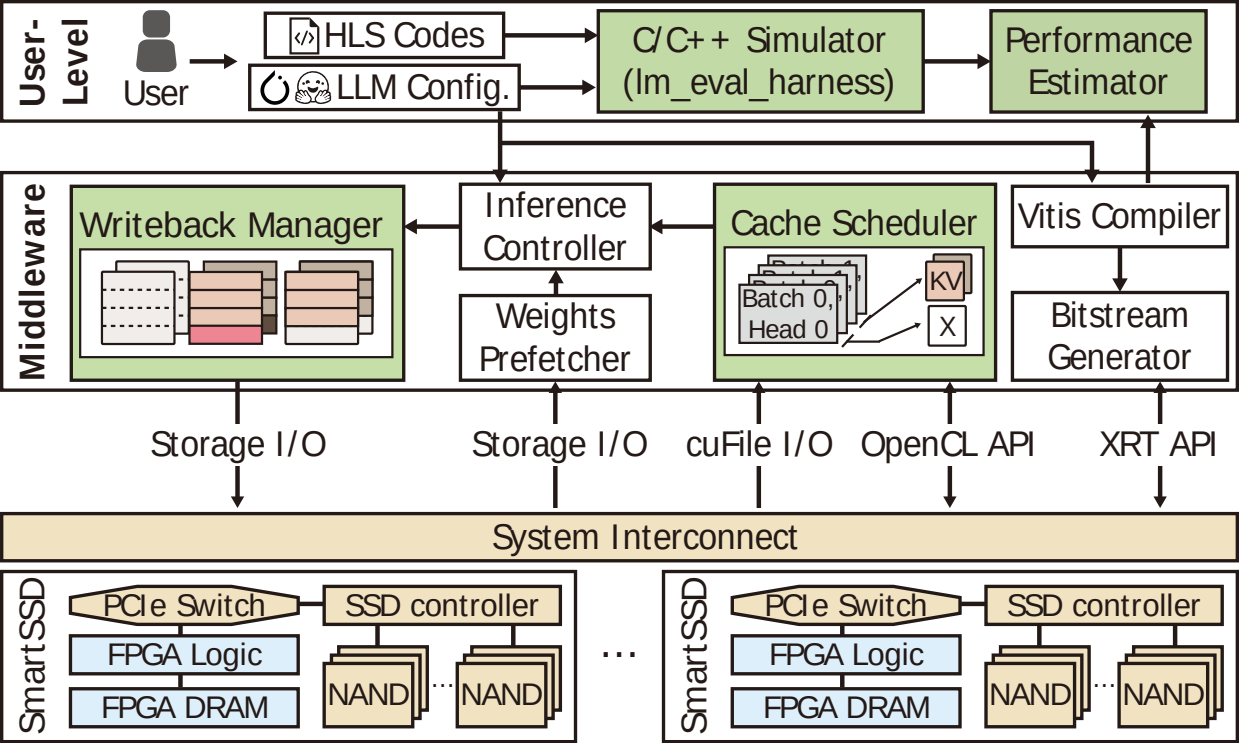
User-Level
User
HLS Codes
LLM Config.
C/C++ Simulator
(lm_eval_harness)
Performance Estimator
Middleware
Writeback Manager
Inference Controller
Weights Prefetcher
Cache Scheduler
Batch 0, Head 0
KV
X
Vitis Compiler
Bitstream Generator
Storage I/O
Storage I/O
cuFile I/O
OpenCL API
XRT API
System Interconnect
SmartSSD
PCIe Switch
SSD controller
FPGA Logic
FPGA DRAM
NAND
NAND
...
SmartSSD
PCIe Switch
SSD controller
FPGA Logic
FPGA DRAM
NAND
NAND